

Calculating and presenting absolute measures of effect: Directly calculating risk differences

There are scenarios in which deriving the risk difference from a relative effect and a baseline risk is problematic:

- i) When the incidence of a binary outcome is low, it is possible to have a large relative effect despite a

small or trivial absolute difference. Consider a study in which adverse events due to treatment occur in 1 in 1,000 patients in the comparator group and 3 in 1,000 in the treatment group. The relative risk of 3 appears large but the absolute effect of 2 in 1,000 more is extremely small.

Similarly, consider a situation in which deaths occur in 3 in 1,000 patients in the comparator group and in 1 in 1,000 patients in the intervention group. The relative risk is 0.33 representing a 67% or 2/3 relative

risk reduction, a large relative effect. The absolute effect is however very small, only 2 in 1,000.

The result of this phenomenon is that one may see inflated estimates of absolute effect when one applies these large relative risks to external baseline risks that are higher than the incidence rates in the available studies (see Example 2 and to a lesser extent Example 1 below).

- ii) The small number of events will typically yield very wide confidence intervals around the point estimate of relative risk. These confidence intervals are likely to be very asymmetrical. The reason is that one boundary of the confidence interval can go only from 1 to 0. The other boundary can be a very large number – 10, 50 or even over 100. The result may be a problematically asymmetric confidence interval with the side greater than 1 much larger than is appropriate (see all three examples below).

The most vivid example of the inappropriate asymmetry is when the point estimate is near a relative risk of 1 and one side of the confidence interval on the risk difference suggests a small benefit or harm and

the other a very large benefit or harm (see example 3 below). Another consequence of the inappropriately wide confidence interval, rating down for imprecision twice instead of once and thus reducing certainty more than is warranted, occurs in all three examples.

- iii) Trials may have 0 events in one or both arms. With 0 events in both arms the results are uninformative with respect to relative effects but very informative with respect to absolute effects: with 0 in both arms the risk difference is 0, the same risk difference one would see whenever the proportion of events is the same in intervention and control groups.

When there are 0 events in both groups whether to leave these trials out of the meta-analysis of relative effects or use a continuity correction remains controversial. Both approaches have limitations.

The solution to all these problems is to abandon our usual approach to estimation of absolute effects by applying baseline risk to relative effect estimates from a meta-analysis. Rather, GRADE users can conduct a meta-analysis of risk differences directly and use the results of this analysis to inform their pooled estimate of the risk difference and the associated 95% confidence interval.

In each example that follows we present first the usual approach applying the relative effects to the baseline risk and then the direct meta-analysis of risk differences. All three examples come from a World Health Organization guideline regarding the management of Arbovirus infections.¹

Example 1

A systematic review considered the effects of platelet transfusion in patients with thrombocytopenia due to dengue virus addressed severe adverse events.¹ The authors identified two studies reporting on severe adverse events with a total of 460 patients and produced a relative risk of 4.63 (95% CI 0.78 to 27.4) from the meta-analysis. The baseline risk in the control group (no platelet transfusion) was 6 per 1000. Using the relative risk and baseline risk, the modeled risk difference was calculated as 22 more per 1000, with a 95% CI from 1 fewer to 158 more per 1000. Due to the wide confidence interval, the certainty of the evidence was rated down twice for imprecision.

Given the asymmetry of the modeled absolute effect's confidence interval, the authors conducted a meta-analysis of risk differences. This produced a risk difference of 15 more per 1000 with a 95% CI of 2 fewer to 32 more per 1000, leading to a more appropriate confidence interval requiring rating down only once for imprecision.

Outcome Timeframe	Study results and measurements	Absolute effect estimates		Certainty of the Evidence (Quality of evidence)	Summary
		No platelet transfusio n	Platelet transfusio n		
Severe adverse events (applying relative effect to baseline risk)	Relative risk: 4.63 (CI 95% 0.78 - 27.4) Based on data from 460 participants in 2 studies Follow up 10.5 days	6 per 1000	28 per 1000	Very low Due to serious risk of bias ¹ and to very serious imprecision. ²	We are uncertain whether platelet transfusion increases severe adverse events
		Difference: 22 more per 1000 (CI 95% 1 fewer - 158 more)			
Severe adverse events (from meta- analysis of risk differences)	Relative risk: 4.63 (CI 95% 0.78 - 27.4) Based on data from 460 participants in 2 studies Follow up 10.5 days	6 per 1000	23 per 1000	Low Due to serious risk of bias ¹ and serious imprecision. ³	Platelet transfusions possibly have little to no increase in serious adverse events
		Difference: 15 more per 1000 (CI 95% 2 fewer - 32 more)			

¹ Risk of Bias: serious. Inadequate/lack of blinding of participants and personnel, resulting in potential for performance bias, Inadequate/lack of blinding of outcome assessors, resulting in potential for detection bias.

² Imprecision: very serious. 95% CI including absence and very large harms.

³ Imprecision: serious. 95% CI including absence and modest harms.

Example 2

The same systematic review also assessed mortality as an outcome.¹ The meta-analysis, which included two studies and 460 patients, found a relative risk of 2.89 (95%CI 0.27 to 31.4). The baseline risk in the control group (no platelet transfusion) was 20 per 1000. Using this relative effect and baseline risk, the

authors calculated an absolute effect of 38 more deaths per 1000, with a 95% CI from 15 fewer to 608 more per 1000. The extremely wide confidence interval warranted a rating down of three levels for imprecision.

Considering the problematically asymmetrical confidence interval, the authors performed a meta-analysis of risk differences, yielding a risk difference of 11 more deaths per 1000, with a 95%CI from 13 fewer to 34 more per 1000. This resulted in a more symmetrical confidence interval but still required downgrading the certainty twice for imprecision.

Outcome Timeframe	Study results and measurements	Absolute effect estimates		Certainty of the Evidence (Quality of evidence)	Summary
		No platelet transfusio n	Platelet transfusion		
Mortality (applying relative effect to baseline risk)	Relative risk: 2.89 (CI 95% 0.27 - 31.4) Based on data from 460 participants in 2 studies	20 per 1000	58 per 1000	Very low Due to extremely serious imprecision. ¹	We are uncertain whether platelet transfusion increases or decreases mortality.
		Difference: 38 more per 1000 (CI 95% 15 fewer - 608 more)			
Mortality (from meta- analysis of risk differences)	Relative risk: 2.89 (CI 95% 0.27 - 31.4) Based on data from 460 participants in 2 studies	20 per 1000	31 per 1000	Low Due to very serious imprecision. ²	Platelet transfusion possibly increases mortality.
		Difference: 11 more per 1000 (CI 95% 13 fewer - 34 more)			

¹ Imprecision: extremely serious, 95% CI including small benefits and very large harms. 2 events in total.
² Imprecision: very serious, 95% CI including modest benefits and harms. 2 events in total.

Example 3

The systematic review also considered organ failure as an outcome,¹ using data from one study with 372 patients. The meta-analysis reported a relative risk of 0.97 (95% CI 0.14 to 6.84). The baseline risk in the control group (no platelet transfusion) was 50 per 1000. Applying the relative effect and baseline risk, the calculated risk difference was 1 fewer per 1000, with 95%CI from 43 fewer to 292 more per 1000. This wide confidence interval led to a two-level downgrade for imprecision.

Given the asymmetrical confidence interval, the authors pooled the risk difference directly. The result was a risk difference of 1 fewer per 1000, with a 95%CI from 21 fewer to 21 more per 1000, requiring only one rating down for imprecision due to the more symmetrical confidence interval.

Outcome Timeframe	Study results and measurements	Absolute effect estimates		Certainty of the Evidence (Quality of evidence)	Summary
		No platelet transfusion	Platelet transfusion		

Organ failure (applying relative effect to baseline risk)	Relative risk: 0.97 (CI 95% 0.14 - 6.84) Based on data from 372 participants in 1 study	50 per 1000	49 per 1000	Very low Due to serious risk of bias ¹ and to very serious imprecision. ²	We are uncertain whether platelet transfusion increases or decreases organ failure.
		Difference: 1 fewer per 1000 (CI 95% 43 fewer - 292 more)			
	Follow up 21 days				
Organ failure (from meta-analysis of risk differences)	Relative risk: 0.97 (CI 95% 0.14 - 6.84) Based on data from 372 participants in 1 study Follow up 21 days	50 per 1000	49 per 1000	Low Due to serious risk of bias ¹ and serious imprecision. ³	Platelet transfusion possibly has little or no impact on organ failure.
		Difference: 1 fewer per 1000 (CI 95% 21 fewer - 21 more)			

¹ Risk of Bias: serious. Inadequate/lack of blinding of participants and personnel, resulting in potential for performance bias, Inadequate/lack of blinding of outcome assessors, resulting in potential for detection bias.

² Imprecision: very serious. 95% CI small and large harms. 11 events.

³ Imprecision: serious, 95% CI including modest benefits and harms. 11 events.

Reference

1. WHO. WHO guidelines for the clinical management and diagnosis of dengue, chikungunya, Zika, and yellow Fever.
<https://www.who.int/publications/i/item/9789275124871>